From Excluded to Consistent: A Vacuum-Tied MOND Law With Per-Host, Condensed-Sourced Settling Against Raw KiDS-1000 Cosmic Shear

Working note

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Abstract

We test a gravity framework in which the acceleration scale is tied to dark energy, $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ ($\kappa = \frac{1}{2}$, fitted), and a cold, collisionless component ("cold energy", $\Omega_c/\Omega_b = 5.364$) settles inside bound systems into the profile the law demands.

The test is framework-native. We run the framework's own 1024^3 particle-mesh simulations in a $100\,h^{-1}\,$Mpc box and score the gravitating-field matter power spectrum on the *raw* KiDS-1000 and DES Y3 shear correlation functions $\xi_\pm$. The comparison is a same-pipeline control run without settling. No ΛCDM-calibrated nonlinear machinery enters the verdict.

The original settling rule, in which all baryons source the law and supply is pooled across overlapping catchments, puts 19–25% too much power at $k = 0.3$–$1\,h\,{\rm Mpc}^{-1}$ and is excluded: $\Delta\chi^2_{\rm KiDS} = +68.7$. Limiting the settling source to a census cap of condensed baryons gives +14.0, still narrowly excluded. Two physically motivated rules together bring the framework's own boost down to 1.005/1.012/1.023 at $k = 0.3/0.5/1$:

- each halo settles only its *own* catchment's supply (no pooling);
- only *condensed* baryons (stars plus cold gas) source the settled profile.

With these rules KiDS-1000 is **consistent**: $\Delta\chi^2 = +2.03$, robustness maximum +3.28, sensitivity tooth passes. The absolute fit, $\chi^2 = 146.3$ ($p = 0.011$), matches the control's 144.3 ($p = 0.014$). DES Y3 is not diagnostic at this box size.

The passing combination was selected after small-box previews. The primary pre-declared combination, which adds the framework's own gas expulsion, is running. The initial conditions remain inherited (ΛCDM-fitted); a native-input rerun is prepared and its preview is unchanged. The alternative footing is pending. We state these limits plainly.

1 Framework and what is assumed

- **Law.** $g = \nu(g_N/a_0)\,g_N$ with the repaired monotone kernel. The saturating-vacuum form $g_N = g^2/(g + a_0)$ gives the same verdicts to within 0.01 (CFG658).
- **Acceleration scale.** $a_0 = \kappa c\sqrt{G\rho_{\rm DE}} \simeq 0.173\,cH_\infty$, with $\kappa = \frac{1}{2}$ fitted. Two footings, 9.3603×10^{-11} and $1.1312 \times 10^{-10}\,{\rm m\,s^{-2}}$, are never pooled.
- **Cold energy.** Its amount is measured, not derived. It is required: without it, clusters, the Bullet Cluster, Local Group timing and cosmic shear all fail (CFG624).
- **Settling.** Inside bound systems, cold energy drifts into the law's "phantom" profile, drawn from each system's turnaround catchment. The switch reads baryons.
- **Provenance.** All numbers come from committed lanes (identifiers CFG n) of the programme's public repository.

2 The test

Simulations. We use a lean copy of the programme's census engine:

- 1024^3 particles, $L = 100\,h^{-1}\,$Mpc, 150 steps from $z_i = 49$;
- snapshots at $z = 1$, 0.755, 0.5, 0.256 and 0;
- the same seed for every run.

Statistic. We use the framework's own boost $B(k, z) = P_{\rm grav}/P_{S0}$ on the *gravitating* field (particles plus settled source; CFG555), applied to the control spectrum. We project it to $\xi_\pm$ with the published redshift distributions and covariances of KiDS-1000 and DES Y3. Points are kept only if at least 90% of their integrand lies inside the box's k-range. Intrinsic alignment is free.

Classes. EXCLUDED if $\Delta\chi^2 \geq 9$ and every robustness variant gives ≥ 9. The robustness variants are: widened covariance, no intrinsic alignment, held-out bins, and DESI distances. A sensitivity tooth ("T-OWN") must show that the framework's own boost, injected into the control, is detectable. Otherwise the result is NOT DIAGNOSTIC.

Freezing. Criteria were committed before each production run. MUTATE checks confirm the null ($F \equiv S0 \Rightarrow \Delta\chi^2 = 0$).

3 Results

rule	B at $k = 0.3/0.5/1$	KiDS $\Delta\chi^2$	robust (max / min)	verdict
all baryons, pooled supply (CFG596)	1.186/1.252/1.206	+68.7	min +43.9	EXCLUDED
condensed census cap (CFG636)	1.058/1.056/1.056	+14.0	min +9.19	EXCLUDED (narrowly)
per-host + condensed (T12)	**1.005/1.012/1.023**	**+2.03**	max +3.28	**CONSISTENT**

For T12:

- KiDS keeps 104 of 225 points. Absolute χ^2 is 146.29 ($p = 0.011$) against 144.26 for the control ($p = 0.014$).
- DES keeps 52 of 227 points: $\Delta\chi^2 = +0.26$, but the tooth fails, so it is NOT DIAGNOSTIC.
- Benchmark: $\sigma_{8,\mathrm{grav}}$ ratio 1.0018.
- Swapping the scorer's inherited background for the framework's own shifts the stored $\Delta\chi^2$ values by ≤ 1.4 (CFG654). The tension was real physics, not an input artefact.

Where the original excess came from.

- 56–79% of CFG596's KiDS $\Delta\chi^2$ sits at $k < 1$, mostly 0.3–1, i.e. the halo scale of groups and clusters (CFG646).
- The engine itself conserved mass and never exceeded supply. But the original rule pooled supply across merged catchments, so dominant hosts drew their neighbours' cold energy: about 10% of the settled mass (CFG647).
- The phantom of hot gas added settled mass where most baryons in massive halos are hot (CFG634).
- The two rules remove exactly these two effects.

4 Limitations (plainly)

1. **Post-preview selection.** The per-host + condensed combination was chosen after seeing 256^3 previews (CFG650 addendum A1). The pre-declared primary run (T1x, which adds gas expulsion) is in progress.
2. **Relative classifier.** The classes compare the framework with a same-pipeline control. Both are marginal in absolute terms ($p \approx 0.01$), so CONSISTENT means "as good as the control", not "a perfect fit".
3. **Inherited initial conditions.** The runs use Planck-fitted parameters and ΛCDM growth to set the starting state. A framework-native-input rerun is validated at 256^3, with an unchanged preview, but has not been run at 1024^3 (CFG654).
4. **Box size.** The box loses 54% (KiDS) and 77% (DES) of points; no baryonic feedback is modelled in either arm; the alternative footing is pending.
5. **Unexplained source rule.** The condensed-only source is not yet derived from an action. The one-action form of the law responds to all baryons (CFG658/722).
6. **Open failure.** Massive systems carry extra core mass not explained by the law: early types +0.28 dex at $Z \approx 3.7$; eRASS1 cluster cores $\times 1.8$ at 7σ (CFG669/687). Two candidate mechanisms failed (CFG721/726).
7. κ is fitted and Ω_c/Ω_b is measured. Nothing here closes the theory.

5 Related results in the programme

Galaxy rotation curves (SPARC, 0.100 dex); wide binaries Newtonian under free-fall basins plus stiffness (-0.8σ against Gaia DR3); cluster satellites; outer cluster lensing on the law ($A = 1.01 \pm 0.09$, 318 eRASS1 clusters);

the primary CMB is untouched. The companion note PAPER47 (DOI 10.5281/zenodo.23354043) describes the one-action form and the nature of the cold-energy carrier. Sealed upcoming tests: a flat $a_0(z)$ (ALMA, 2026-11-06 and 11-28) and Gaia DR4 wide binaries and Milky Way force (2026-12-02).

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